Marie Sklodowska-Curie Postdoctoral Fellowships 2025 Call
Candidate Expression of Interest form

This Fellowship Candidate EOI must be **limited to 3 sides of A4**
Please keep answers brief and within the stated word limits.

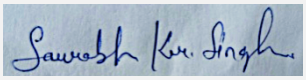
Please email your completed form along with your **two-page CV** and **full publications list** to: fellowships@swansea.ac.uk by **5pm on 11th April 2025 labelled "MSCA PF 2025 EOI"**

Applicant Name:	SAURABH KUMAR SINGH
Email Address:	s.k.singh@swansea.ac.uk
Current Organisation & Role (If applicable):	Swansea University (Postdoctoral Research Assistant)
Swansea University Host Supervisor identified and agreed with both parties: **Named Supervisor:**	Yes Dr. Helen Chadwick
Swansea University Host Faculty/Research institute Research in the faculties - Swansea University	Faculty of Science and Engineering
Postdoctoral Fellowship Call type **Application stage**	☑ Standard European Fellowship ☐ Global Fellowship ☑ First submission 2025 ☐ Resubmission (2022/2023/2024-delete as appropriate) ☐ Previous score _____%

Eligibility Declaration: I can confirm that I can evidence that I:

- Am a postdoctoral researcher in possession of a doctoral degree (defined as a successfully defended doctoral thesis)
- Have **a maximum** of 8 years full time equivalent research experience (measured from the date of PhD award)
- Have **not** resided or carried out my main activity (work, studies etc) in the UK for more than 12 months in the 3 years immediately before the call deadline (**10th September 2025**)

I confirm that if I am given the opportunity to submit my proposal, I will not withdraw from the process unless circumstances beyond my control prevent me from doing so.

Signature: 	**Date: 07/04/2025**

PART A: To be completed by the candidate

Field of Specialisation* *Enter details of your field(s) of specialisation. (20 words max.)*
Molecule surface dynamics, state-resolved scattering, energy transfer, rotational orientation-controlled reactivity, and gas-surface interactions under ultra-high vacuum conditions.

Personal statement*/Candidate profile:
My fascination with molecular interactions at surfaces led me to pursue a PhD at TIFR Hyderabad, where I investigated CO_2 dissociation on Cu(110) and developed quantum state-resolved surface scattering techniques. This work provided me with expertise in surface preparation, UHV characterization, and laser-based rovibrational state preparation, deepening my understanding of gas-surface reactivity at solid interfaces.

To further explore molecular orientation effects, I joined Swansea University's Surface Dynamics Lab as a postdoc under Dr. Helen Chadwick, studying rotational orientation-controlled reactive scattering of H_2 on metal surfaces. While my research has primarily focused on reactivity, I now seek to expand into energy exchange in non-reactive scattering, particularly the role of surface phonons in gas-surface interactions.
The MSCA Fellowship will be instrumental in my transition to an independent research career focused on molecule-surface dynamics. By integrating reactive and non-reactive scattering, I aim to advance our understanding of energy dissipation at surfaces, with applications in astrochemistry, catalysis, and energy conversion. The fellowship will provide essential training, international collaborations, grant-writing expertise, and leadership experience, equipping me to establish my own research group and make meaningful contributions to the field.

Research & Training Proposal:

Understanding energy exchange at molecule-surface interfaces is fundamental to surface chemistry. Surface phonon excitation plays a crucial role in reactive sticking, influencing molecular adsorption, diffusion, and dissociation. At low incident energies, molecules often follow a precursor-mediated pathway, temporarily trapping before dissociation, yet phonon-mediated energy transfer mechanisms remain underexplored.
Helium spin echo has been instrumental in probing atomic-scale energy exchange at surfaces with unprecedented resolution. Recent advancements extend this concept to molecular systems, enabling quantum state-resolved studies of molecule-surface interactions. The magnetic molecular interferometer (MMI) (Nat. Commun., 8, 15357 (2017)) enables precise nuclear and rotational spin control, achieving ultra-high energy resolution in surface phonon measurements (Phys. Chem. Chem. Phys., 24, 14198-14208 (2022)). This breakthrough opens new possibilities for probing energy transfer at molecule-surface interfaces, advancing our understanding of phonon contributions to reactive systems. Building on this progress, this proposal aims to systematically investigate phonon-mediated energy transfer in molecule-surface scattering, using both chemically inert (^3He) and reactive (D_2) probes.

Research Objectives

1. **Investigating energy dissipation through surface phonons using chemically inert (^3He) and reactive probes (D_2)**
   Using ^3He as an inert probe, a reference for phonon excitation and energy dissipation will be established without reactivity. Introducing D_2 under identical conditions will reveal differences in energy transfer mechanisms between reactive and non-reactive probes.
2. **Surface Temperature-Dependent Energy Transfer and phonon excitation**
   By varying surface temperature while keeping incident energy constant, I will determine how thermally excited phonons influence energy dissipation.
3. **Incident Energy Effects on Phonon Excitation**
   With constant surface temperature and varying incident energy, I will study how translational energy affects phonon excitation, impacting reaction probability and surface energy dynamics.
4. **Molecular Orientation Dependence Energy transfer and phonon excitation**
   Using rotational orientation control, I will explore how molecular alignment influences energy transfer, offering insights into anisotropic gas-surface interactions.

This study aims to systematically explore phonon-mediated energy transfer in molecular scattering, offering insights into energy dissipation at interfaces and potentially revealing the role of surface phonons in chemical reactivity.

Excellence

This project employs the MMI molecular beam technique to make the first detailed energy exchange measurements through surface phonons in molecular systems. It integrates surface physics, quantum dynamics, and molecular scattering, with implications for catalysis, energy conversion, astrochemistry, and materials

science. The research could reveal how energy exchange through surface phonons influences adsorption and reaction probabilities.

Impact
This research will provide benchmark data to refine theoretical models of energy transfer, impacting surface science and astrochemistry. Insights into energy dissipation during molecule formation, particularly in dust and ice environments, will be valuable. Findings will be shared via open-access publications and major conferences to ensure broad visibility.

Implementation
Experiments will be conducted at Swansea University's Surface Dynamics Lab, home to the only molecular beam spin-echo (MMI) facility. Proof-of-concept studies confirm the feasibility of phonon detection via molecular beam techniques. Supervised by Dr. Helen Chadwick, the project will enable two-way knowledge transfer in molecular beam experiments.

Training & Career Development
This fellowship provides hands-on training in operating the novel MMI setup, UHV systems, cryostats, and MATLAB-based data analysis. It also fosters professional growth in leadership, project management, scientific writing, and research ethics through conferences and publications.

Why have you chosen SU as your Host Institution for this fellowship (100 words max)
Swansea University's Surface Dynamics Lab hosts the only molecular beam spin-echo (MMI) facility capable of studying molecule-surface scattering, unlike other setups limited to atomic systems. This project is made possible by the technique and analysis developed by Dr. Helen Chadwick and Prof. Gil Alexandrowicz, whose expertise in quantum state-resolved molecule surface scattering is globally recognized. Swansea's interdisciplinary approach, cutting-edge facilities, and commitment to surface science innovation provide the ideal environment for advancing my research and career development through the MSCA Postdoctoral Fellowship.

PART B: Host Endorsement: to be completed by the Host Supervisor

Saurabh will be supervised by Dr Helen Chadwick and mentored by Prof. Gil Alexandrowicz. Both have experience of developing successful research proposals for highly competitive schemes, with Helen being awarded a Future Leaders Fellowship (FLF), and Gil obtaining both a European Research Council Early Career and Consolidator grant. During the development stage, Helen will offer advice on the structure and content of the proposal, as well as reading and commenting on all aspects. She will also make her FLF Case for Support available.
During the project, both Helen and Gil will teach and support Saurabh with regards to learning how to use the apparatus and perform the data analysis to successfully complete the project. They will also highlight training courses, internal and external career development opportunities (e.g., those offered by the Learned Society of Wales Researcher Development Programme) and relevant conferences to Saurabh and encourage him to engage in at least 10 days professional development in line with the Researcher Development Concordat.
The Faculty will be required to provide administrative support in finalising the budget for the proposal, and grant management support if successful. The lab and office space Saurabh will require to successfully complete the project is already available.

Host Supervisor Signature:	**Date:**

Saurabh Kumar Singh

s.k.singh@swansea.ac.uk | Phone: +91 9005119516
ORCID: 0000-0002-2760-9671
Surface Dynamics Lab, Swansea University, Wales, United Kingdom

Research Profile

My current research focuses on exploring alignment- and orientation-controlled reactive scattering processes in gas-surface chemical reactions on metal surfaces, with particular emphasis on the H_2 helicopter and cartwheel motion. I have developed expertise in quantum state-resolved surface scattering studies and high-resolution molecular laser spectroscopy.

Education

Postdoctoral Research Assistant Oct 2024 – Present
Surface Dynamics Lab, Swansea University, Wales, UK
Supervisor: Dr. Helen Chadwick

Ph.D. in Chemical Science 2018 – 2024
Tata Institute of Fundamental Research (TIFR), Hyderabad, India
Supervisor: Dr. Pranav R. Shirhatti
Thesis: *Understanding the Dissociation Dynamics of CO_2 on Cu(110) and Developing Techniques for State-Selected Molecule-Surface Scattering Experiments*

M.Sc. in Physical Chemistry 2016 – 2018
Banaras Hindu University, Varanasi, India
CGPA: 8.58/10

B.Sc. (Hons.) in Chemistry 2012 – 2015
Banaras Hindu University, Varanasi, India
CGPA: 7.7/10

Research Experience

Postdoctoral Research Assistant, Swansea University
Surface Dynamics Lab
Project: Rotationally controlled H_2 dissociation on single-crystal metal surfaces.

- Magnetic manipulation techniques to control the preparation of specific rotational nuclear spin quantum states in a molecular beam.
- Investigating alignment and orientation-dependent reactivity in gas-surface interactions.

Ph.D. Researcher, TIFR Hyderabad
Thesis: Dissociation dynamics of CO_2 on Cu(110); development of experimental techniques.

- Performed molecular beam scattering experiments to study molecule-surface reactions.
- Developed novel methods for quantum state-selected scattering experiments.
- Constructed UHV-compatible laser diagnostic setups for surface reaction studies.

Technical Skills

- **Experimental Techniques:** Molecular beam scattering, surface science (UHV systems), helium atom scattering, velocity map imaging (VMI), photoacoustic spectroscopy.
- **Instrumentation Development:** Laser frequency stabilization, ion/spatial imaging, non-linear laser optics, high-finesse optical cavity design, high-precision laser spectroscopy.
- **Software and Simulation:** MATLAB, LabVIEW, Python, SIMION (ion trajectory simulations), MolFlow+ (vacuum system simulations).
- **Analytical Techniques:** Laser spectroscopy (single- and multi-photon excitation), vacuum technology, mass spectrometry.

Fellowships and Awards

- Nominated for the Schmidt Science Fellows 2025 through TIFR Hyderabad (2024)
- Selected participant at the Global Young Scientists Summit (GYSS), 17–20 January 2023, Singapore.
- Junior Research Fellowship (CSIR-JRF), Government of India (2017)

Conferences / Workshops

- Poster on *"Dissociation Dynamics of CO on Cu(110) Surface"* at iCOMET-2023, November 12–17, 2023.
- Poster on *"Dissociation Dynamics of CO on Cu(110) Surface"* at SSDS-2023, October 5–8, 2023.
- Poster on *"Design and Development of Diagnostic Tools for Quantum State-Specific Studies"* at SDMC-2022, November 10–13, 2022.
- Poster on *"Design and Development of Diagnostic Tools for Quantum State-Specific Studies"* at TIFR Hyderabad In-house Symposium, September 15–17, 2022.
- Poster on *"Design and Development of Low-Cost Wavemeter"* at TACC 2020, March 03–05, 2021.

References

- **Prof. Gil Alexandrowicz**
  Professor, Chemistry, Swansea University, Wales, United Kingdom
  Email: g.n.alexandrowicz@swansea.ac.uk
- **Dr. Helen Chadwick**
  Senior Lecturer, Chemistry, Swansea University, Wales, United Kingdom
  Email: h.j.chadwick@swansea.ac.uk

Publications & Patents

Saurabh Kumar Singh

Patents

1. **Saurabh Kumar Singh**, Avinash Kumar, and Pranav R. Shirhatti. *A multistage thermal drift compensated cross-fringed Fizeau interferometer wavemeter and method thereof.* **Patent:** Indian App. No. 202321048766, filed on 20-Jul-2023, TIFR.

Publications

Preprint URLs available at: https://orcid.org/0000-0002-2760-9671

1. **Saurabh Kumar Singh**, and Pranav R. Shirhatti. *Dissociation Dynamics of CO_2 on Cu(110) Studied over a Wide Range of Incident Energies.* **The Journal of Physical Chemistry C 128, 49**, 20841–20848 (2024).
2. **Saurabh Kumar Singh**, and Pranav R. Shirhatti. *The Curious Case of CO_2 Dissociation on Cu(110).* **The Journal of Chemical Physics 160.2**, 024702 (2024).
3. **Saurabh Kumar Singh**, Avinash Kumar, and Pranav R. Shirhatti. *A Simple and Low-Cost Setup for Part-Per-Billion Level Frequency Stabilization and Characterization of Red He-Ne Laser.* **HardwareX 14**, e00421 (2023).
4. Geetika Bhardwaj, **Saurabh Kumar Singh**, and Pranav R. Shirhatti. *A Compact and Highly Collimated Atomic/Molecular Beam Source.* **Review of Scientific Instruments 94.4** (2023).
5. Avinash Kumar, **Saurabh Kumar Singh**, and Pranav R. Shirhatti. *A Versatile and Narrow Linewidth Infrared Radiation Source for Ro-Vibration State-Selected Preparation of Molecules in Molecular Beams.* **Review of Scientific Instruments 94 (11)** (2023).

Manuscripts Under Preparation

1. **Saurabh Kumar Singh**, Avinash Kumar, and Pranav R. Shirhatti. *Design and Characterization of a High-Performance Wavemeter Based on Crossed Fringe Fizeau Interferometer (CFFI).*
2. Amaraja Taur, **Saurabh Kumar Singh**, and Pranav R. Shirhatti. *Identifying Signatures of Thermal and Non-Thermal Reaction Pathways in Plasmon-Induced H_2 + D_2 Exchange Reaction.* ChemRxiv (2021). *doi:10.26434/chemrxiv-2021-87t1c.*